

Radeon Company Ltd. — Enterprise Networking Project #5

Implementation Documentation | Cisco Packet Tracer

Table of Contents

1. Project Overview
 2. Network Topology Design
 3. Device Inventory
 4. IP Addressing & Subnetting Plan
 5. VLAN Assignment Plan
 6. Step-by-Step Configuration Guide
 - 6.1 Basic Configuration — L2 Switches
 - 6.2 Basic Configuration & SSH — L3 Switches & Routers
 - 6.3 VLAN & Port Security — L2 Switches
 - 6.4 Trunk Port Configuration
 - 6.5 IP Addressing — Routers & L3 Switches
 - 6.6 OSPF Configuration
 - 6.7 DHCP Server Configuration
 - 6.8 InterVLAN Routing & DHCP Helper
 7. Known Issues & Recommended Fixes
 8. Testing & Verification Checklist
-

1. Project Overview

Client: Radeon Company Ltd. (Banking & Insurance) **Location:** Nairobi, Kenya (African branch) **Building:** 4-story office building **Simulation Tool:** Cisco Packet Tracer

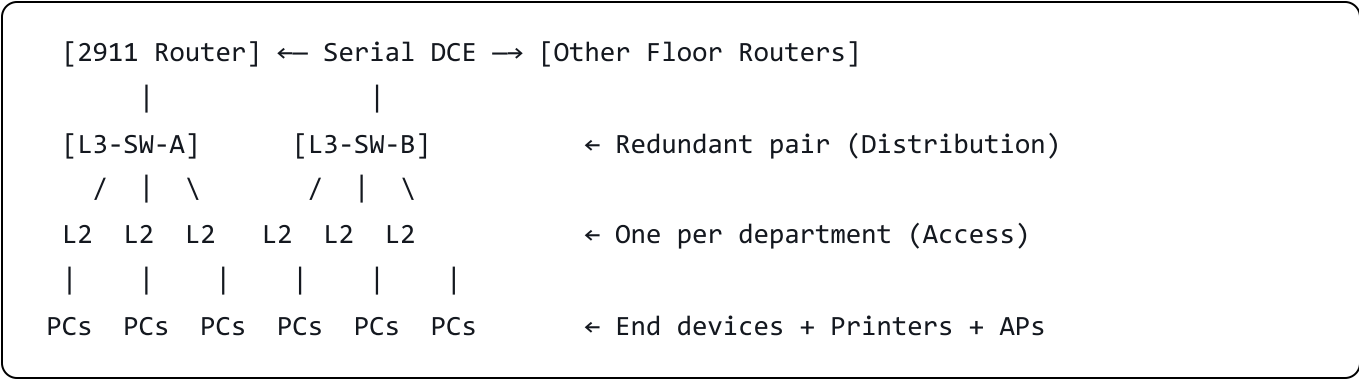
Radeon Company Ltd. requires a fully functional enterprise network for its first African branch. The network must support multiple departments across four floors, provide wired and wireless connectivity, enforce security policies, and be resilient to single points of failure through a hierarchical redundant design.

2. Network Topology Design

The network follows the **three-layer hierarchical model**:

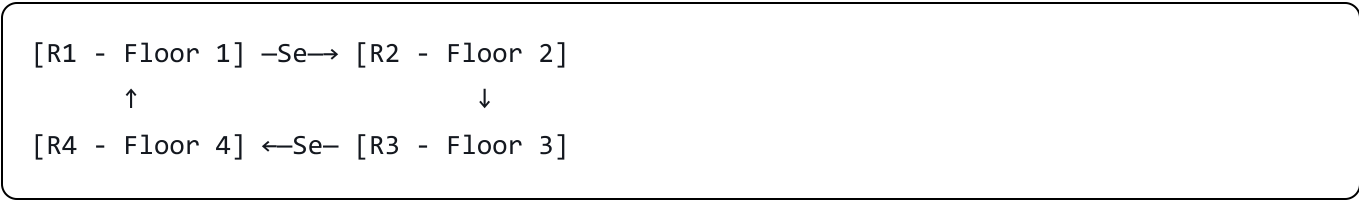
Layer	Devices Used	Purpose
Core	Cisco 2911 Routers (1 per floor)	Inter-floor routing via Serial DCE links
Distribution	Cisco 3650 L3 Switches (2 per floor)	InterVLAN routing, redundancy, DHCP relay
Access	Cisco 2960 L2 Switches (1 per department)	End-device connectivity, port security

Per-Floor Topology



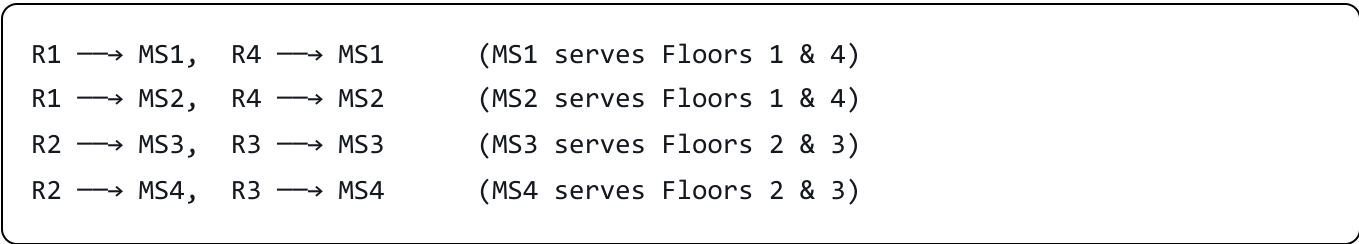
Both L3 switches connect to ALL L2 switches on their floor to ensure redundancy. If one L3 switch fails, the other maintains connectivity.

Inter-Router Serial Connections (Core Layer — Ring Topology)



The four routers form a **ring** (R1↔R2, R2↔R3, R3↔R4, R4↔R1) via Serial DCE links, providing a redundant path in the core layer.

Router ↔ L3 Distribution Switch Connections



Each pair of multilayer switches (MS1/MS2 and MS3/MS4) is connected to two routers for redundancy, ensuring continuity if one router fails.

OSPF is used across all routers and L3 switches to share routing information dynamically.

3. Device Inventory

Per Floor (×4 floors, adjusted for departments)

Qty	Device	Model	Role
1	Router	Cisco 2911	Core routing
2	L3 Switch	Cisco 3650	Distribution (redundant pair)
3	L2 Switch	Cisco 2960	Access (one per department)
1+	Access Point	Generic AP	Wireless per department
—	End PCs & Printers	—	Per department quantities

Fourth Floor — Additional Devices

Qty	Device	Purpose
1	DHCP Server	Assign IPs to all departments
1	HTTP Server	Internal web services
1	Email Server	Internal email services
2	Admin PCs	Server Room management

4. IP Addressing & Subnetting Plan

Base Network: 192.168.1.0 Subnet Size for All Departments: /26 → 62 usable hosts (accommodates ~60 users) Subnet Size for Router Serial & Switch Links: /30 → 2 usable hosts (point-to-point)

4.1 Department VLANs — Subnetting Table

#	Floor	Department	VLAN	Network	Subnet Mask	First Host	Last Host
1	Floor 1	Management	10	192.168.1.0	255.255.255.192	192.168.1.1	192.168.1.62
2	Floor 1	Research	20	192.168.1.64	255.255.255.192	192.168.1.65	192.168.1.126
3	Floor 1	Human Resource	30	192.168.1.128	255.255.255.192	192.168.1.129	192.168.1.190
4	Floor 2	Marketing	40	192.168.1.192	255.255.255.192	192.168.1.193	192.168.1.254
5	Floor 2	Accounts	50	192.168.2.0	255.255.255.192	192.168.2.1	192.168.2.62
6	Floor 2	Finance	60	192.168.2.64	255.255.255.192	192.168.2.65	192.168.2.126
7	Floor 3	Logistics & Store	70	192.168.2.128	255.255.255.192	192.168.2.129	192.168.2.190
8	Floor 3	Customer Care	80	192.168.2.192	255.255.255.192	192.168.2.193	192.168.2.254
9	Floor 3	Guest Area	90	192.168.3.0	255.255.255.192	192.168.3.1	192.168.3.62
10	Floor 4	Administration	100	192.168.3.64	255.255.255.192	192.168.3.65	192.168.3.126
11	Floor 4	ICT	110	192.168.3.128	255.255.255.192	192.168.3.129	192.168.3.190
12	Floor 4	Server Room	120	192.168.3.192	255.255.255.192	192.168.3.193	192.168.3.254

4.2 Router Serial Links — Ring Topology (Point-to-Point /30)

Link	Network	Subnet Mask	R-A Interface IP	R-B Interface IP	Broadcast
R1 ↔ R2 (serial)	10.10.10.0	255.255.255.252	10.10.10.1	10.10.10.2	10.10.10.3

Link	Network	Subnet Mask	R-A Interface IP	R-B Interface IP	Broadcast
R2 ↔ R3 (serial)	10.10.10.4	255.255.255.252	10.10.10.5	10.10.10.6	10.10.10.7
R3 ↔ R4 (serial)	10.10.10.8	255.255.255.252	10.10.10.9	10.10.10.10	10.10.10.11
R4 ↔ R1 (serial)	10.10.10.12	255.255.255.252	10.10.10.13	10.10.10.14	10.10.10.15

4.3 Router ↔ L3 Switch Links (Point-to-Point /30)

Link	Network	Subnet Mask	Router IP	L3 Switch IP	Broadcast
R1 ↔ MS1	10.10.10.16	255.255.255.252	10.10.10.17	10.10.10.18	10.10.10.19
R4 ↔ MS1	10.10.10.20	255.255.255.252	10.10.10.21	10.10.10.22	10.10.10.23
R4 ↔ MS2	10.10.10.24	255.255.255.252	10.10.10.25	10.10.10.26	10.10.10.27
R1 ↔ MS2	10.10.10.28	255.255.255.252	10.10.10.29	10.10.10.30	10.10.10.31
R2 ↔ MS3	10.10.10.32	255.255.255.252	10.10.10.33	10.10.10.34	10.10.10.35
R3 ↔ MS3	10.10.10.36	255.255.255.252	10.10.10.37	10.10.10.38	10.10.10.39
R3 ↔ MS4	10.10.10.40	255.255.255.252	10.10.10.41	10.10.10.42	10.10.10.43
R2 ↔ MS4	10.10.10.44	255.255.255.252	10.10.10.45	10.10.10.46	10.10.10.47

Note: MS1 & MS2 are the redundant distribution switch pair for Floors 1 & 4 (connected to both R1 and R4). MS3 & MS4 serve Floors 2 & 3 (connected to both R2 and R3).

5. VLAN Assignment Plan

VLAN ID	Department	Floor	Network	Gateway (SVI)	Wildcard Mask
10	Management	1	192.168.1.0/26	192.168.1.1	0.0.0.63
20	Research	1	192.168.1.64/26	192.168.1.65	0.0.0.63
30	Human Resource	1	192.168.1.128/26	192.168.1.129	0.0.0.63
40	Marketing	2	192.168.1.192/26	192.168.1.193	0.0.0.63

VLAN ID	Department	Floor	Network	Gateway (SVI)	Wildcard Mask
50	Accounts	2	192.168.2.0/26	192.168.2.1	0.0.0.63
60	Finance	2	192.168.2.64/26	192.168.2.65	0.0.0.63
70	Logistics & Store	3	192.168.2.128/26	192.168.2.129	0.0.0.63
80	Customer Care	3	192.168.2.192/26	192.168.2.193	0.0.0.63
90	Guest Area	3	192.168.3.0/26	192.168.3.1	0.0.0.63
100	Administration	4	192.168.3.64/26	192.168.3.65	0.0.0.63
110	ICT	4	192.168.3.128/26	192.168.3.129	0.0.0.63
120	Server Room	4	192.168.3.192/26	192.168.3.193	0.0.0.63

Wildcard Mask = 255.255.255.255 – Subnet Mask Example: 255.255.255.255 – 255.255.255.192 = **0.0.0.63** The **Gateway (SVI)** column is the IP you assign to `interface` `vlan [ID]` on the L3 switches.

6. Step-by-Step Configuration Guide

6.1 Basic Configuration — L2 Switches

Repeat this for every L2 (2960) switch. Replace `[name]` and `[password]` with actual values.

```
> en
# conf t
# hostname [DeptName-L2-SW]
# banner motd # Authorized Access Only #
# no ip domain-lookup
# enable secret [password]
# service password-encryption
# line console 0
# password [password]
# login
# exit
# line vty 0 15
# password [password]
# login
# exit
# do wr
```

6.2 Basic Configuration & SSH — L3 Switches and Routers

SSH must be configured on **all routers and L3 switches** as per project requirements.

```
> en
# conf t
# hostname [DeviceName]
# banner motd # Authorized Access Only #
# no ip domain-lookup
# enable secret [password]
# service password-encryption
# ip domain-name radeon.com
# username admin password [password]
# crypto key generate rsa
    → Enter 1024 when prompted for modulus size
# line console 0
# password [password]
# login
# exit
# line vty 0 15
# login local
# transport input ssh
# exit
# do wr
```

6.3 VLAN & Port Security — L2 Switches

Configure the access ports connected to end devices. Check which ports are actually connected before using the range command.

```
> en
# conf t
# vlan [VLAN_ID]
# name [DepartmentName]
# exit
# int range fa0/2-24
# switchport mode access
# switchport access vlan [VLAN_ID]
# switchport port-security
# switchport port-security maximum 2
# switchport port-security mac-address sticky
# switchport port-security violation shutdown
# exit
# do wr
```

`maximum 2` allows one PC and one printer per port. Adjust as needed per port.

6.4 Trunk Port Configuration — L2 Switches

The uplink port(s) from each L2 switch going to L3 switches must be in trunk mode to carry all VLANs.

```
> en
# conf t
# int range fa0/1-2
# switchport mode trunk
# exit
# do wr
```

Check physical connections and update the interface range accordingly. On the L3 switch side, the downlink ports to L2 switches must also be set to trunk mode.

6.5 IP Addressing — Routers & L3 Switches

On each Router — Serial interfaces (DCE side requires clock rate):

```
> en
# conf t
# int se0/0/0
# ip address [IP] [SubnetMask]
# clock rate 64000          ← Only on DCE end
# no shutdown
# exit
# do wr
```

On L3 Switches — Convert port to Layer 3 before assigning IP:

```
> en
# conf t
# int gig0/0
# no switchport
# ip address [IP] [SubnetMask]
# no shutdown
# exit
# ip routing
# do wr
```

`no switchport` converts a Layer 2 port to a Layer 3 routed port. This must be done before assigning an IP address on the L3 switch physical interfaces.

6.6 OSPF Configuration

On Routers:

```
> en
# conf t
# router ospf 1
# network [NetworkAddress] [WildcardMask] area 0
# ← Repeat for every directly connected network
# passive-interface [interface] ← Apply on all user-facing interfaces
# exit
# do wr
```

Example for R1 (Floor 1 Router):

```
# router ospf 1
# network 10.10.10.0 0.0.0.3 area 0      ← R1-R2 serial link
# network 10.10.10.12 0.0.0.3 area 0    ← R4-R1 serial link
# network 10.10.10.16 0.0.0.3 area 0    ← R1-MS1 link
# network 10.10.10.28 0.0.0.3 area 0    ← R1-MS2 link
# exit
```

On L3 Switches (enable IP routing first):

```
> en
# conf t
# ip routing
# router ospf 1
# network [NetworkAddress] [WildcardMask] area 0
# ← List all VLAN SVI networks and router-facing link networks
# exit
# do wr
```

6.7 DHCP Server Configuration

The DHCP Server is located in the **Server Room on the Fourth Floor**. Configure it through the Services tab in Packet Tracer.

1. Click the DHCP Server → **Services** tab → **DHCP**
2. Create one pool per department:

Pool Name	Default Gateway	DNS Server	Start IP	Subnet Mask
VLAN10-Pool	192.168.1.1	192.168.3.193	192.168.1.2	255.255.255.192
VLAN20-Pool	192.168.1.65	192.168.3.193	192.168.1.66	255.255.255.192
VLAN30-Pool	192.168.1.129	192.168.3.193	192.168.1.130	255.255.255.192
VLAN40-Pool	192.168.1.193	192.168.3.193	192.168.1.194	255.255.255.192
VLAN50-Pool	192.168.2.1	192.168.3.193	192.168.2.2	255.255.255.192
VLAN60-Pool	192.168.2.65	192.168.3.193	192.168.2.66	255.255.255.192
VLAN70-Pool	192.168.2.129	192.168.3.193	192.168.2.130	255.255.255.192
VLAN80-Pool	192.168.2.193	192.168.3.193	192.168.2.194	255.255.255.192
VLAN90-Pool	192.168.3.1	192.168.3.193	192.168.3.2	255.255.255.192
VLAN100-Pool	192.168.3.65	192.168.3.193	192.168.3.66	255.255.255.192
VLAN110-Pool	192.168.3.129	192.168.3.193	192.168.3.130	255.255.255.192

Assign static IPs manually to the 3 servers and 2 Admin PCs in the Server Room (192.168.3.193–192.168.3.197 recommended). The DHCP server itself uses 192.168.3.193.

6.8 InterVLAN Routing & DHCP Helper — L3 Switches

Both L3 switches on each floor must have the same VLANs created and the same SVI IP addresses and DHCP helper addresses configured.

Step 1 — Create VLANs on L3 Switch:

```
> en
# conf t
# vlan 10
# vlan 20
# vlan 30
  ← Continue for all VLANs relevant to this floor
# exit
```

Step 2 — Assign SVI IPs and DHCP Helper:

```
# int vlan 10
# ip address 192.168.1.1 255.255.255.192
# ip helper-address 192.168.3.193    ← IP of the DHCP server
# no shutdown
# exit
# int vlan 20
# ip address 192.168.1.65 255.255.255.192
# ip helper-address 192.168.3.193
# no shutdown
# exit
# int vlan 30
# ip address 192.168.1.129 255.255.255.192
# ip helper-address 192.168.3.193
# no shutdown
# exit
    ← Continue for all VLANs (40 through 120) using addresses from Section 5
# do wr
```

Important: The `ip helper-address` command forwards DHCP broadcast requests from devices to the centralized DHCP server in the server room. Without this, devices will not receive IPs automatically.

7. Known Issues & Recommended Fixes

The following issues were identified in the original configuration steps. Each should be corrected before or during implementation.

7.1 Typos in CLI Commands

#	Location	Incorrect	Correct
1	Step 5	<code>#conf that</code>	<code>#conf t</code>
2	Step 3	<code>#siwtchport port-security violation shutdown</code>	<code>#switchport port-security violation shutdown</code>
3	Step 5 (wildcard note)	<code>255.255.255.255.192</code>	<code>255.255.255.192</code>

7.2 Missing Commands

#	Location	Issue	Fix
4	Step 4 & 5	<code>no shutdown</code> missing after IP assignment	Add <code>no shutdown</code> on every interface after assigning IP
5	Step 4	<code>clock rate</code> missing on DCE serial interfaces	Add <code>clock rate 64000</code> on the DCE end of all serial links
6	Step 3	Port-security maximum has no value	Change to <code>switchport port-security maximum 2</code> (or 1 as needed)
7	Step 5	OSPF process ID missing	Change <code>router ospf</code> → <code>router ospf 1</code>
8	Step 6.8	<code>no shutdown</code> missing on VLAN SVIs	Add <code>no shutdown</code> after assigning each SVI IP
9	Step 7	VLANs not created on L3 switch before SVI assignment	Create VLANs with <code>vlan [ID]</code> before entering <code>int vlan [ID]</code>

7.3 Security Issues

#	Issue	Fix
10	<code>enable password</code> is used — stores password in plaintext	Replace with <code>enable secret [password]</code> which uses MD5 hashing
11	SSH not configured on 2911 Routers (Step 2.2 only covers L3 switches)	Apply the full SSH configuration block (Step 6.2) to all 4 routers as required

7.4 Design & Logic Issues

#	Issue	Recommendation
12	Step 6 says "go to IT department DHCP server"	DHCP server is in the Server Room on Floor 4, not the IT department

#	Issue	Recommendation
13	VLAN list in Step 7 only goes up to VLAN 60	The network has 12 departments — VLANs must go up to 120 (10 through 120 in steps of 10)
14	No redundancy protocol for L3 switch pair	Implement HSRP on VLAN SVIs so one switch is Active and the other is Standby. Without HSRP, both switches will have the same default gateway IP causing conflicts and OSPF router-ID collisions
15	No OSPF passive interfaces configured	Add <code>passive-interface</code> for all interfaces facing end devices to prevent unnecessary OSPF hellos on user ports
16	No trunk VLAN filtering on L3 switch uplinks	Use <code>switchport trunk allowed vlan [list]</code> to restrict VLANs on each trunk and reduce broadcast traffic
17	Guest Area has 40 PCs + wireless	With 40 PCs, 2 printers, and wireless users, the total could exceed 62. Monitor actual user count; upgrade to /25 (126 hosts) if needed
18	Server Room VLAN SVI acts as default gateway	Manually configure static IPs on all 3 servers pointing to the Server Room SVI as their default gateway

8. Testing & Verification Checklist

Use the following commands to verify the network is working correctly:

On Routers / L3 Switches

# show ip interface brief	→ Check all interfaces are up/up
# show ip route	→ Verify OSPF routes are learned
# show ip ospf neighbor	→ Verify OSPF adjacencies formed
# show vlan brief	→ Verify VLANs exist
# show interfaces trunk	→ Verify trunk ports and VLANs allowed
# show ip dhcp binding	→ View assigned IPs from DHCP
# show port-security interface fa0/x	→ Verify port security per port
# show running-config	→ Review full configuration

End Device Tests (Packet Tracer PDU / CLI Ping)

Test	Expected Result
PC in VLAN 10 pings PC in VLAN 10 (same floor)	✓ Success
PC in VLAN 10 pings PC in VLAN 40 (different floor)	✓ Success via OSPF
PC receives IP automatically via DHCP	✓ IP from correct pool
PC accesses HTTP Server via browser	✓ Web page loads
PC sends email via Email Server	✓ Email delivered
SSH from Admin PC to Router	✓ SSH login successful
Unauthorized device on access port triggers violation	✓ Port shuts down

Appendix: Wildcard Mask Reference

Subnet Mask	Wildcard Mask	Usable Hosts	Used For
255.255.255.192 (/26)	0.0.0.63	62	All 12 department VLANs
255.255.255.252 (/30)	0.0.0.3	2	All router serial & switch links

*Document prepared for Radeon Company Ltd. — Enterprise Networking Project #5
Simulation: Cisco Packet Tracer / Protocol: OSPF / Security: SSH + Port-Security*